

Advanced Theory — Module 07: Communication —

Study Questions

1. How does multi-agent Active Inference differ from single-agent? What changes when the environment includes other agents?
2. What is recursive modeling? How deep can the recursion go (Theory of Mind levels)?
3. How does communication lead to shared generative models? What is the mechanism?
4. How do cultural norms function as shared generative models?
5. What does consensus mean in free energy terms? When is collective free energy minimized?
6. How is Nash equilibrium recast in terms of expected free energy?
7. When does cooperation emerge in Active Inference? What must be in agents' C vectors?
8. How is signaling formalized? How does a sender choose signals?
9. What is generalized synchrony between coupled agents? How does it relate to "being on the same wavelength"?
10. How does Active Inference connect to dynamical systems theory of coordination (Haken-Kelso)?
11. What is stigmergy? How does it implement indirect communication?
12. How does swarm intelligence emerge from individual free energy minimization?
13. How do social institutions implement collective belief updating?
14. What is cultural evolution as model selection?
15. How does the depth of recursive modeling affect social competence?
16. What happens when agents' generative models are fundamentally incompatible?
17. How does Active Inference game theory handle mixed-motive games (Prisoner's Dilemma)?
18. What is the "common knowledge" problem? How does it relate to shared priors?
19. How does trust between agents relate to precision weighting of social observations?
20. Critically evaluate: Can Active Inference explain the full complexity of human social behavior, or does it oversimplify?